

LAB 1 – Install and Test

● Install simpleSSD-standalone in **ubuntu 20.04**

- Official Website Link: [simpleSSD](https://simplessd.github.io/)
- Lab Instruction Link: [LAB1 - hackmd](#)

● Lab requirements

➤ 題目敘述

本次作業需要同學安裝 simpleSSD-standalone 並嘗試執行。

接著請錄製 Demo 影片：展示執行的過程，以及簡單解釋幾個問題。

➤ 範例結果

```
s98420@labtest:~/simplessd-standalone$ ./simplessd-standalone ./config/sample.cfg
g ./simplessd/config/sample.cfg ./output
SimpleSSD Standalone v2.0
***** Begin of simulation *****
*** Statistics of Request Generator ***
Tick: 61298946200
Time (ps): 0 - 61298946200 (61298946200)
I/O (bytes): 16777216 (273695015.005005 B/s)
I/O (counts): 4096 (Read: 4096, Write: 0)
*** End of statistics ***
*** Statistics of Block I/O Entry ***
Latency (ns): min=4972.580000, max=9874.420000, avg=4975.267344, stdev=157369.57
2170
*** End of statistics ***
*** Statistics of Event Engine ***
Simulation Tick (ps): 61298946200
Host time duration (sec): 15.409757
Event handled: 12294 (797.806215 ops)
*** End of statistics ***
End of simulation @ tick 61298946200
```

IO Generator的資訊

IO操作的最低延遲 / 最高延遲 / 平均延遲 / 標準差

Event Engine

➤ 評分標準

- 1 (60%) 在 Demo 影片裡執行一次 simpleSSD
- 2 (10%) 在 [simpleSSD-standalone/simplessd/config/sample.cfg](#) 裡找到下圖的片段，分別使用 **SLC** (Single Level Cell) 和 **MLC** (Multi Level Cell) 運行 simpleSSD，並截圖範例結果裡平均延遲的差別

▼ 需要更改的部分

```
253 ## Set type of NAND flash
254 # Possible values:
255 # 0: Single Level Cell    設定值為0~2分別代表
256 # 1: Multi Level Cell    SLC、MLC、TLC
257 # 2: Triple Level Cell
258 # Please check NAND timing has valid values
259 # SLC: LSB timing should valid
260 # MLC: LSB and MSB timing should valid
261 # TLC: LSB, CSB and MSB timing should valid
262 NANDType = 1
```

3 (10%) 承上題，解釋導致延遲差距的原因

4 (10%) 在 [simpleSSD-standalone/config/sample.cfg](#) 裡找到下圖的片段，將 `readwrite` 模式改為 `randrw`，`rwmixread` 設為 `0.2`，`io_size` 分別用 `128 MB` 和 `4096 MB` 運行一次，截圖範例結果裡最大延遲的差別

▼ 需要更改的部分

```
107 ## Total size to I/O = int
108 # Only valid when time_based = false 所有IO加起來的大小
109 io_size = 128M # 128 * 2^20 bytes
110
111 ## Type of I/O = str 要產生什麼樣的IO
112 # Possible values:
113 # read: Sequential read
114 # write: Sequential write
115 # randread: Random read
116 # randwrite: Random write
117 # readwrite: Mixed sequential I/O
118 # randrw: Mixed random I/O
119 readwrite = write
120
121 ## Percentage of mixed I/O = float
122 # If readwrite is not readwrite nor randrw, this value will ignored
123 # 0 <= value <= 1 讀取在全部IO裡佔多少比例
124 rwmixread = 0.5 # 50% of read I/O
```

5 (10%) 承上題，當 `io_size` 為 `4096 MB` 時，最大延遲會在甚麼情況下出現

➤ 繳交格式

hw1/

└─ hw1_2.jpg -> 第二題的截圖

└─ hw1_4.jpg -> 第四題的截圖

└─ hw1_video.mp4 -> 執行 simpleSSD 及第三、五題的解釋

➤ 繳交期限

10/6 23:59

繳交區會多開放一個禮拜至 10/13 為止，超過繳交期限就算遲交，遲交一天扣本次作業 5 分，如果到 10/13 才繳交總共會-35 分